

Problem Set 8 — Third Law & Friction

Physics 2026–27 · due at the start of class, Monday, November 2

NAME _____

DATE _____

How to work this set. Friction template, every time: find N from vertical equilibrium (it equals mg *only* when nothing else pushes vertically), then $f_s \leq \mu_s N$ or $f_k = \mu_k N$, then $\Sigma F = ma$. Remember the $\leq$: static friction is a *ceiling*, not a fixed value.

1 Name the reaction

THIRD LAW

For each action force, name the reaction: magnitude, direction, and — most importantly — *which object it acts on*.

- A swimmer's hand pushes water backward.
- Earth's gravity pulls down on a falling mango.
- A hammer strikes a nail, pushing it downward.

2 Why don't they cancel?

THIRD LAW

A classmate argues: "*If every force has an equal and opposite reaction, the forces always cancel, so nothing should ever accelerate.*" In two or three sentences, find the flaw. (The key word is **objects**.)

ANSWER

3 Two coefficients, two numbers

FRICTION

A crate weighing **300 N** sits on a level floor with $\mu_s = 0.40$ and $\mu_k = 0.30$. (Note: you're given the *weight*, not the mass.)

- Find the maximum static friction force.
- Find the kinetic friction force once it's sliding.
- Which is bigger, and what does that mean physically about starting vs. keeping a slide going?

WORK

4 Does it move?

STATIC FRICTION

A **15 kg** toolbox sits on a floor with $\mu_s = 0.45$. You push horizontally with **60 N**.

- Run KDEA: find $f_{s,\max}$ and compare it to your push. Does the toolbox move?
- Trick part: what is the actual friction force acting on the toolbox right now? (Careful — it is *not* $f_{s,\max}$.)

K	EVERY GIVEN, WITH UNITS
D	THE TARGET & ITS UNIT
E	N = MG HERE — SAY WHY THAT'S ALLOWED · $F_{s,\max} = M_s N$
A	SYMBOLS FIRST · UNITS THE WHOLE WAY

5 Sliding with friction

KINETIC FRICTION

A **30 kg** equipment case is already sliding across the gym floor ($\mu_k = 0.25$) while being pushed horizontally with **120 N**. Draw the FBD, then run KDEA for the acceleration.

K	EVERY GIVEN, WITH UNITS
D	THE TARGET & ITS UNIT
E	N, THEN $F_k = M_k N$, THEN $\Sigma F = F_a - F_k = MA$
A	SYMBOLS FIRST · UNITS THE WHOLE WAY

6 The angled pull

CHALLENGE

A 10 kg box is dragged across a floor ($\mu_k = 0.20$) by a rope pulling with 40 N at 30° above horizontal.

- Find N . (The rope's upward component is holding part of the weight — N is *less* than mg .)
- Find the friction force.
- Find the acceleration.
- One sentence: why did pulling at an angle change the friction, when nothing about the floor changed?

K	EVERY GIVEN, WITH UNITS · DECOMPOSE T FIRST
D	THE TARGETS & THEIR UNITS
E	VERTICAL: $N = MG - T \sin\theta$ · HORIZONTAL: $\Sigma F = T \cos\theta - \mu_k N = MA$
A	SYMBOLS FIRST · UNITS THE WHOLE WAY

Graded on the homework 4-3-2-1 scale. Problem 4b is the week's trap: static friction supplies exactly what's needed to balance the push — the ceiling number is only the breaking point.